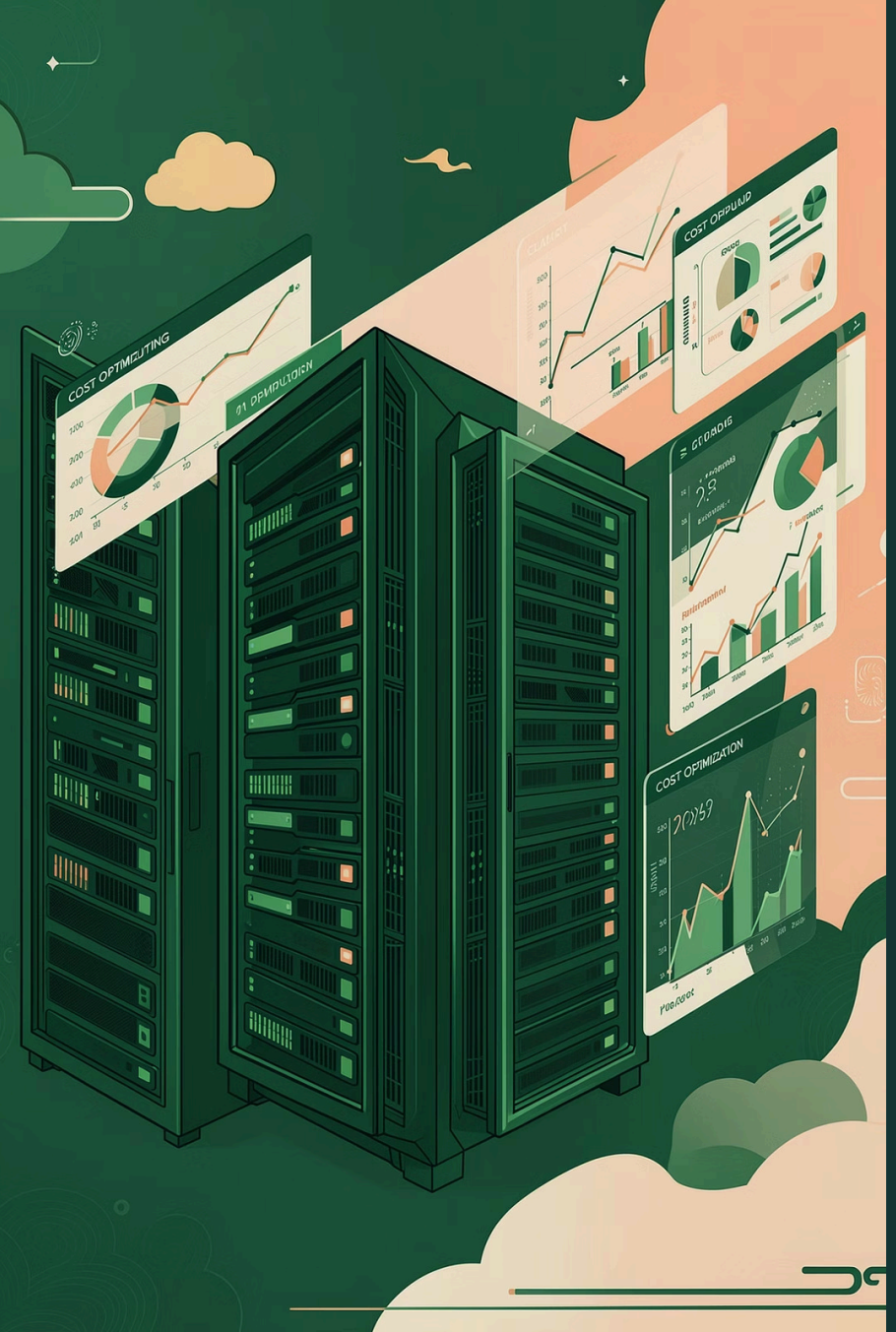
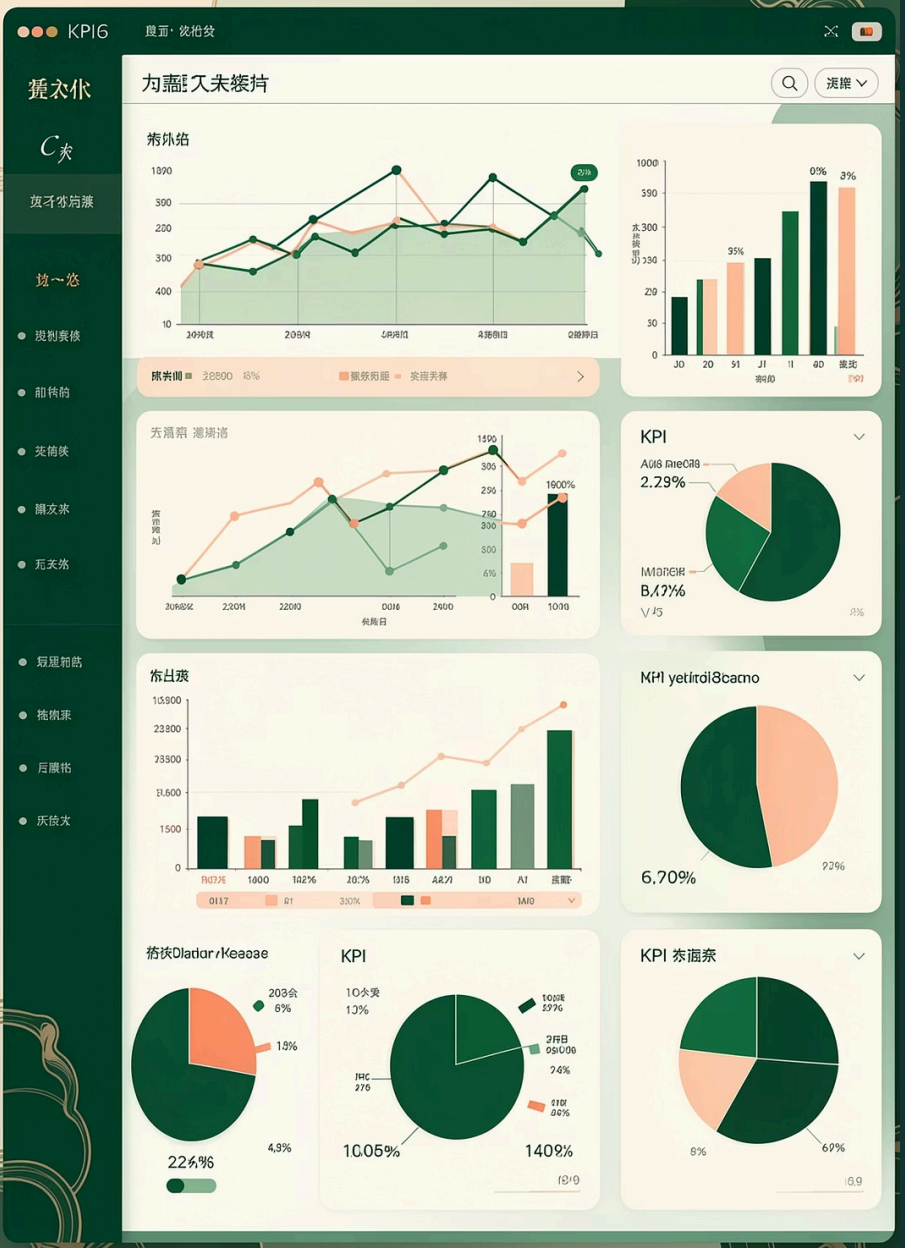


Cloud Cost Analysis & Optimization

Transforming raw cloud usage data into actionable insights to reduce unnecessary expenses and optimize resource allocation.

Made with GAMMA



Project Overview

Cost Inefficiencies

Identify wasted spending across cloud infrastructure

Idle Resources

Detect underutilized servers and instances

Optimization

Generate actionable recommendations for savings

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Dataset Summary

Core Data Fields

- Server ID and Region
- Instance Type specifications
- CPU and Memory Usage
- Hours Running and Cost per Hour

Data Challenges

- Missing values requiring imputation
- Inconsistent formatting issues
- Unstructured data organization
- Derived metrics needed

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Data Cleaning & Structuring

01

Handle Missing Values

Impute or remove incomplete records

02

Standardize Formats

Ensure consistent data structure

03

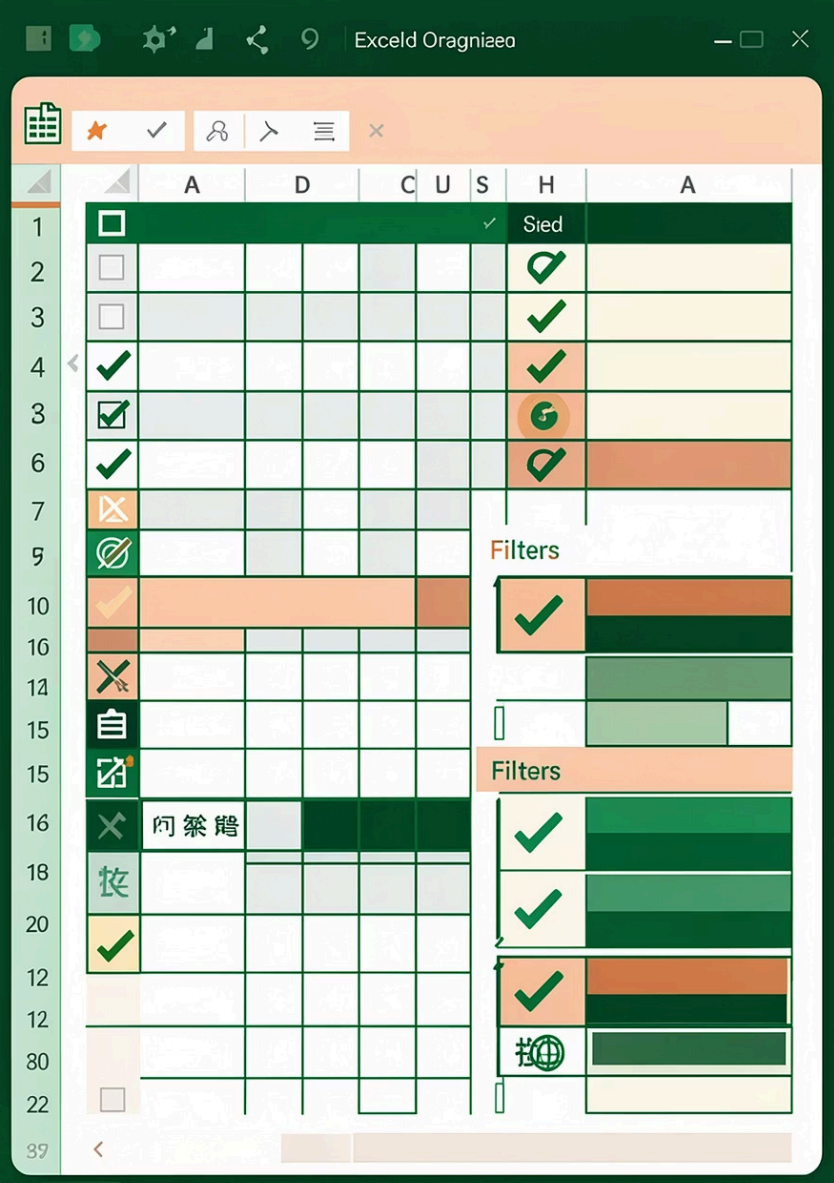
Structure Dataset

Organize for efficient analysis

04

Verify Consistency

Validate data quality across columns



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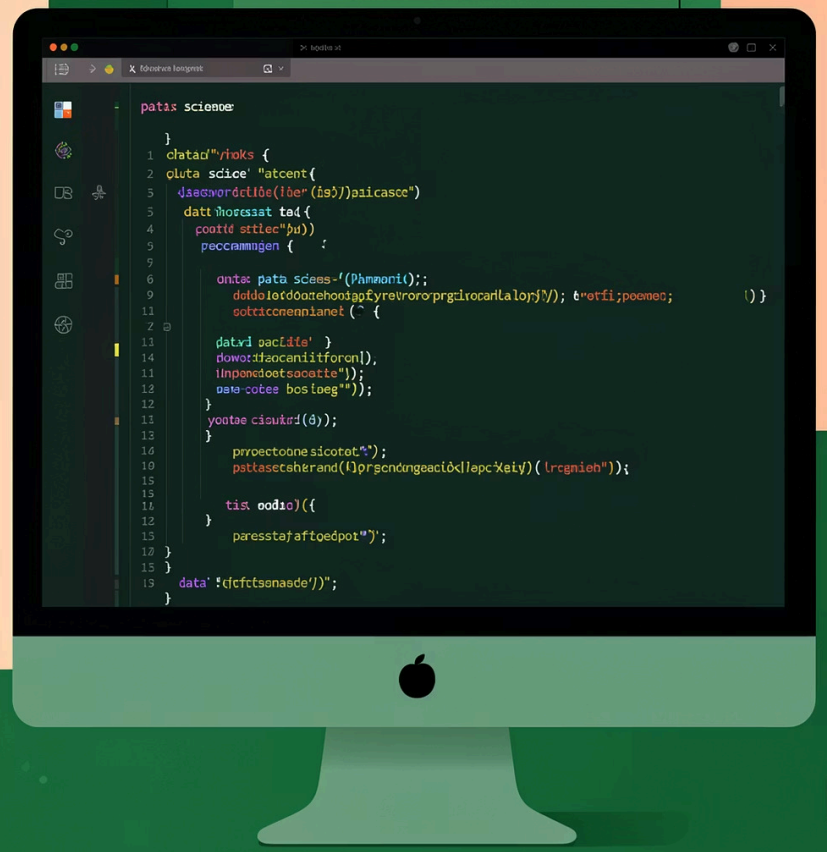

Python Analysis with Google Colab

Data Loading

Import dataset using Pandas library for manipulation

Feature Engineering

Create Daily Cost = Hours Running × Cost per Hour

Status Classification

Label servers as Idle or Active based on CPU and Memory usage

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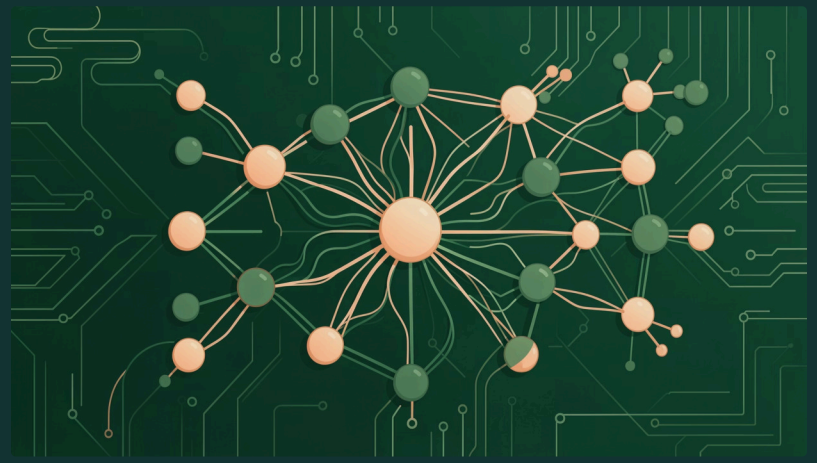
Machine Learning & Recommendations

Linear Regression Model

Predict CPU usage patterns to identify underutilization trends and forecast resource needs.

Generated Actions

- Idle servers → "Shutdown" recommendation
- Active servers → "Keep Running" status
- Calculate total potential savings



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SQL Analysis

Query Total Cost

Calculate aggregate expenses across all servers

Identify Idle Servers

Detect resources generating unnecessary costs

Calculate Savings

Quantify potential cost reduction opportunities

Compare Metrics

Analyze costs by region and instance type

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Power BI Dashboard

Interactive visualizations transform complex data into clear, actionable insights.

KPI Cards

Total Cost, Potential Savings, and Idle Servers Count displayed prominently

Cost Distribution

Breakdown by Region, Instance Type, and server status

Savings Opportunities

Visual identification of optimization areas

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Key Insights

Idle Server Impact

Significant portion of costs attributed to underutilized resources

Regional Variations

Certain regions contribute disproportionately to cloud expenses

Optimization Potential

Strategic identification of idle resources enables major cost reductions

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Real-World Application & Conclusion

Business Value

- Reduce unnecessary cloud spending
- Identify underutilized resources
- Support better cost management
- Enable data-driven decisions

Complete Solution

This end-to-end project demonstrates how cloud cost data can be cleaned, analyzed, and visualized using Excel, Python, SQL, and Power BI to identify inefficiencies and optimize spending.

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